# PALASH KRISHNA VISHWAS

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## SKILLS

- **Programming Languages:** Python | JavaScript | C++
- **Backend Development:** Node.js | Django | Microservices | Systems Design
- **Frontend Development:** React | React Native | Css
- **Databases:** MongoDB | MySQL | ClickHouse | Redis | Kafka | RabbitMQ | Data Modeling

## EXPERIENCE

**Software Developer | Node.js | Python Microservices | ReactJS | MongoDB | Redis | Clickhouse**
Digital Web Solutions, CollabX, Delhi (Remote) *Since Feb 2025*

- **Led the technical direction and execution of CollabX** from ground up, transforming an **MVP into a production-scale system** supporting **50K+ influencers**.
- **Architected and implemented core backend and microservices infrastructure** across finance, influencer lifecycle, content management, campaign orchestration, and RBAC System—designing MongoDB transactional models and building ClickHouse analytics pipelines for high-performance reporting.
- **Designed and developed reusable React modules** while defining application-wide state management, **collaborating closely with product and design teams** to shape key UI and experience decisions across the platform.
- **Partnered closely with Product and Ops** to define feature roadmaps, execution strategy, and operational workflows.

**Full Stack Developer Intern | React | NodeJs | MongoDB**
Wobb, Delhi (Remote) *Jan 2025 - Feb 2025*

- **Resolved production issues** and **optimized backend APIs (indexing, caching, query tuning) to improve performance and scalability** of influencer management and campaign workflows.
- Worked with **Docker, Kubernetes, and Azure/AWS** to support containerized deployments and scalable backend services.

## TOP PROJECTS

**OLAP Database (MergeTree Inspired) | C++ | Columnar Storage | Compaction | Parallel Query Execution** *Feb 2026*

- **Built an analytical-grade columnar OLAP database** inspired by **ClickHouse's MergeTree** architecture to understand analytical engine internals.
- Implemented a custom columnar storage format with compression-aware reads, **GROUP BY aggregation pipeline**, and **parallel query execution** supporting COUNT, SUM, AVG, MIN, and MAX.
- Implemented MergeTree-style compaction and part merging to preserve sorted data layout for efficient analytical queries.

**OLTP Database (InnoDb Inspired) | C++ | B+ Tree | Parser | Buffer Pool | TCP Server | Storage Engine** *Jan 2026*

- **Built a row-oriented OLTP database engine** in C++ with fixed-size slotted-page storage and **clustered B+ Tree indexing** for efficient point and range queries.
- Implemented core components including a **buffer pool** with LRU eviction, SQL-like **query parser**, **catalog manager**, and TCP-based client-server architecture.
- D**eveloped insertion, deletion, and index/range scan execution** using a **volcano-style iterator model** under single-transaction semantics.

**Oasis | Adaptive Learning Platform | Node.js | React | MongoDB | Redis | TrueSkill** *May 2025 - Dec 2025*

- **Architected and developed a full-stack adaptive learning platform for JEE/NEET preparation**, implementing dynamic difficulty adjustment and chapter-level performance modeling to enable personalized mastery-based practice.
- **Designed a real-time quiz engine** using webSockets with session orchestration and probabilistic difficulty calibration powered by a **TrueSkill-based rating system**.
- **Led cross-functional execution** across **engineering**, **design**, **ops**, **marketing**, and **product**—driving technical architecture, product strategy, and end-to-end delivery from concept to launch.

## EDUCATION

Bachelor of Technology, **Kalinga Institute of Industrial Technology** (KIIT), **Bhubaneswar**, **Odisha** (2021–2025)